TRAINING DAY · SESSION 4 · LIBRARY

Questioning for everyone

What do I do with this answer?

1. After asking a question	2. The question is hard, or the class is quiet	3. Choosing who answers	4. Student says "I don't know"	5. Student gives a correct answer	6. Open it to the class
⏳ **Wait time**	👥 **Think-Pair-Share**	🎯 **Cold calling**	🔁 **No opt out**	🔎 **Follow-up questions**	🏀 **ABC: Agree, Build, Challenge**

1 ⏳ AFTER ASKING A QUESTION

Wait time questioning technique

STEPS

1. Ask the whole class. No name yet.
2. Pause 3–5 seconds: count on your fingers, scan the room.
3. Name one student.
4. After the answer, pause about 3 seconds before you respond.

THE SCIENCE

Thinking takes time. Working memory needs a few seconds to find an answer and put it into words, and longer in a second language.

Rowe (1986); Tobin (1987). Teachers typically wait under 1 second. At about 3 seconds or more, answers become longer, more students volunteer, and "I don't know" falls.

Second-language classes. One study found teachers in second-language classes waited even less: about 0.7 seconds after an answer (Shrum, 1985).

2 👥 THE QUESTION IS HARD, OR THE CLASS IS QUIET

Think-Pair-Share questioning technique

STEPS

1. Pose the question.
2. "10 seconds: think on your own." No talking.
3. Pair turns: "Partner A, 20 seconds. Then swap."
4. Cold call: "What did your partner say?"

THE SCIENCE

Peer discussion (Smith et al., 2009). After discussing a question in pairs, students did better on a new question about the same idea, even when nobody in the pair knew the answer at first.

Structure matters. Talk helps learning when it is structured: everyone thinks first, both partners talk, and there is a clear task and time limit (EEF Teaching and Learning Toolkit: collaborative learning).

3 🎯 CHOOSING WHO ANSWERS

Cold calling questioning technique

STEPS

1. Ask the question to everyone.
2. Pause and scan the whole room.
3. Name a student, hand up or not.
4. Thank them, and spread questions around.
5. Nervous student? Warn them: "I'm coming to you next."

THE SCIENCE

Everyone has to think. If anyone might be asked, everyone has to prepare an answer. Answering out loud helps memory most, so ask lots of different students.

Dallimore et al. (2013, 2019). In university classes with frequent cold calling, voluntary participation and comfort rose over the year, and the gender gap in participation closed.

Tone matters (Morek et al., 2022). In middle-school lessons, a warm, conversational cold call built belonging. A demanding "you must answer" style raised anxiety.

4 🔁 STUDENT SAYS "I DON'T KNOW"

No opt out questioning technique

STEPS

1. Ask a question and name a student.
2. Can't answer? A classmate helps, give a cue, or give the answer.
3. Go back to the same student: "Now you tell me..."
4. Finish only when that student gives the right answer.

THE SCIENCE

Try, then get it right. Trying to answer, then getting the right answer, helps memory more than just being told (Kornell, Hays & Bjork, 2009). No opt out ends with the student saying it themselves.

Teacher expectations (Good, 1981). Teachers call on lower attainers less, wait less for them, and don't stay with them after a wrong answer. No opt out deliberately reverses this.

5 🔎 STUDENT GIVES A CORRECT ANSWER

Follow-up questioning technique

STEPS

1. Ask your opening question.
2. Correct answer? Don't say "Good" and move on.
3. Ask the same student at least two follow-ups.
4. Use two different types from the bank.

Clarify "What do you mean by...?"
Evidence "How do you know?"
Assumptions "What are you assuming?"
Viewpoints "What would someone who disagrees say?"
Consequences "If that's true, what follows?"

THE SCIENCE

Elaborative interrogation. Answering "why" and "how" links new knowledge to what students already know (Dunlosky et al., 2013).

Self-explanation (Bisra et al., 2018). Prompting students to explain their thinking improved learning across 64 studies. "How do you know?" is a self-explanation prompt.

6 🏀 OPEN IT TO THE CLASS

ABC (Agree, Build, Challenge) questioning technique

STEPS

1. Ask an open question and take one answer.
2. Keep a neutral face. Don't judge it: bounce it.
3. "Agree, build or challenge? Why?"
4. Aim for 3+ students before you speak again.

THE SCIENCE

Everyone listens. If anyone may be asked to agree, build or challenge, everyone has to listen to each answer ("Accountable Talk").

"Because..." every time. Asking for a reason makes students explain their thinking, which improves learning (Bisra et al., 2018).

Dialogic teaching. An EEF trial in 76 primary schools found about 2 months' extra progress in English and science.

"Memory is the residue of thought." Daniel Willingham. Students learn what they think about. Hands-up questioning lets most of the class skip the thinking; every technique here makes thinking expected of everyone. Rosenshine, Principle 3: "Ask a large number of questions and check the responses of all students."

Much of this evidence is older or from universities, and none of the techniques has been trialled by name. Treat them as strongly supported principles, not proof.